

Dark Matter Module: the physics behind

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1 Dark Matter

- **Target topics:** Dark Matter (galaxy rotation curves, clusters).
- **Learning objectives:** Analyze multiple lines of evidence for dark matter; derive galaxy rotational velocity curve from mechanics, Newton gravitation law and Kepler law; investigate galaxies and clusters composition; find the cluster mass from virial theorem reviewing kinetic energy and gravitational potential; connect classical Physics to Cosmology.
- **Physics program link:** Mechanics, Kepler's law, Newton's law of gravitation, centripetal acceleration, circular motion, centripetal force, gravitational force, energy and momentum conservation, gravitational potential energy, kinetic energy.

1.1 Galaxy rotational velocity curve

- **Objectives:** Apply the law of gravitation to derive the expected orbital velocity in a galaxy. Compare the theoretical velocity with observational data from galaxy datasets. Introduce the dark matter hypothesis as a potential solution to the discrepancy.
- **Dataset variables of galaxies:** [5]
 - Rad (radius)
 - V_{obs} (observed velocity)
 - $\text{err}V$ (velocity error)
 - V_{gas} (gas velocity)
 - V_{disk} (stellar disk velocity)
 - V_{bul} (bulge velocity)

- SB_{disk} (surface brightness disk)
- SB_{bul} (surface brightness bulge)

Classical Physics derivation:

- We model a small mass m (orbiting body) in a stable circular orbit around a large central mass M (e.g., galactic center) at a radius r .
- The condition for stable orbit requires the Gravitational Force (F_G) to equal the Centripetal Force (F_c).

1. Gravitational and Centripetal Forces

- **Gravitational Force (F_G):**

$$F_G = \frac{GMm}{r^2} \quad (1)$$

- **Centripetal Force (F_c):**

$$F_c = \frac{mv^2}{r} \quad (2)$$

- **Equating the Forces ($F_G = F_c$):**

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad (3)$$

2. Deriving the rotational velocity (v)

- **Cancellation of m :** The mass of the orbiting body (m) cancels out, showing v is independent of m .

$$\frac{GM}{r^2} = \frac{v^2}{r} \quad (4)$$

- **Isolating v^2 :** Multiply by r to isolate v^2 .

$$v^2 = \frac{GM}{r} \quad (5)$$

- **Final Rotational Velocity (v):** Take the square root.

$$v = \sqrt{\frac{GM}{r}} \quad (6)$$

- **Physical Meaning:** This Keplerian velocity profile dictates that $v \propto 1/\sqrt{r}$, meaning orbital speed decreases as distance from the central mass increases.

3. Connection to Kepler's third law

- **Orbital Period (T):** Period is the circumference divided by velocity.

$$T = \frac{2\pi r}{v} \quad (7)$$

- **Kepler's relation :** Square T and substitute $v^2 = GM/r$.

$$T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{GM/r} \implies T^2 = \frac{4\pi^2}{GM} r^3 \quad (8)$$

- **Physical Meaning:** This confirms Kepler's Third Law ($T^2 \propto r^3$), demonstrating that the proportionality derived from the force balance is fully consistent with the relationship between orbital period and radius.

Velocity and mass

- **Baryonic velocity (from data):**

$$V_{\text{baryons}} = \sqrt{V_{\text{gas}}^2 + V_{\text{disk}}^2 + V_{\text{bul}}^2} \quad (9)$$

- **Baryonic mass:**

$$M_{\text{baryons}}(r) = \frac{r V_{\text{baryons}}^2(r)}{G} \quad (10)$$

- **Total observed mass:**

$$M_{\text{obs}}(r) = \frac{r V_{\text{obs}}^2(r)}{G} \quad (11)$$

- **Dark matter velocity,** using the velocity components from data:

$$V_{\text{DM}} = \frac{V_{\text{obs}}^2(r)}{r} - \frac{V_{\text{gas}}^2(r)}{r} - \frac{V_{\text{disk}}^2(r)}{r} - \frac{V_{\text{bul}}^2(r)}{r} \quad (12)$$

- **Dark matter density (from data):**

$$\rho_{\text{DM}}(r) = \frac{1}{4\pi r^2} \frac{dM_{\text{DM}}}{dr} = \frac{1}{4\pi G r^2} \frac{d}{dr} (r^2 V_{\text{DM}}^2(r)) \quad (13)$$

NFW Dark Matter Profile [8, 7]

- **NFW profile for dark matter:**

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (14)$$

where r_s is the scale radius and ρ_s is the characteristic density, $\alpha = 1, \beta = 3, \gamma = 1$.

$$r_{200} = \left[\frac{3M_{200}}{4\pi 200\rho_c} \right]^{1/3} \quad (15)$$

with the critical density: $\rho_c = 2.775 \cdot 10^2 M_\odot/\text{kpc}^3$.

- The characteristic radius:

$$r_s = \frac{r_{200}}{C} \quad (16)$$

- The concentration parameter is:

$$C = 0.3 \cdot 11.7 \cdot \left[\frac{M_{200}}{10^{11} M_\odot} \right]^{-0.075} \quad (17)$$

- **Matching observed density and NFW:**

$$\rho_{\text{DM}}(r_{\text{match}}) = \rho_{\text{NFW}}(r_{\text{match}}) \quad (18)$$

From this, obtain M_{200} , r_{200} , and c . The concentration parameter $c = r_{200}/r_s$.

- **NFW dark matter mass:**

$$M_{\text{NFW}}(r) = 4\pi \int \rho_{\text{NFW}} r^2 dr = 4\pi \rho_s r_s^3 \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (19)$$

- **Velocity dark matter:**

$$V_{\text{DM}}(r) = \sqrt{\frac{GM_{\text{NFW}}(r)}{r}} \quad (20)$$

- **Total simulated curve:**

$$V_{\text{sim}}(r, f) = \sqrt{\frac{GM_{\text{baryons}}(r) + f \cdot M_{\text{DM}}(r)}{r}} \quad \text{with } f \in [0, 2] \quad (21)$$

Data visualization and fit

- **Plot:**

– **X-axis:** Rad (radius, data)

– **Y-axis:**

* V_{baryons} (red)

* V_{obs} (data, blue) $\pm \text{errV}$

* V_{sim} (green)

- **Fit χ^2/dof (chi-squared):**

$$\chi^2/\text{dof} = \frac{1}{N - P} \sum_{i=1}^N \left(\frac{V_{\text{obs}}(r_i) - V_{\text{sim}}(r_i)}{\text{errV}(r_i)} \right)^2 \quad (22)$$

where N is the number of data points and P is the number of free parameters.

Detailed explanation from literature [3, 4, 5, 9, 6]

1) Centrifugal Equilibrium

$$\frac{V_{\text{obs}}^2(r)}{r} = a_H(r) + a_D(r) + a_{HI}(r) \quad (23)$$

where:

- $a_H(r)$ = radial acceleration due to DM halo
- $a_D(r)$ = radial acceleration due to stellar disk
- $a_{HI}(r)$ = radial acceleration due to gas disk

For a spherical halo (Eq. 8, [4]):

$$a_H(r) = \frac{4\pi G}{r^2} \int_0^r \rho_H(r') r'^2 dr' \quad (24)$$

2) Analytical derivation

Multiplying Eq. (23) by r^2 (Eq. 9 [4]):

$$r^2 V^2(r) = 4\pi G \int_0^r \rho_H(r') r'^2 dr' + r^2 (a_D(r) + a_{HI}(r)) \quad (25)$$

Taking the derivative with respect to r gives the DM halo density:

$$\rho_H(r) = \frac{1}{4\pi G r^2} \frac{d}{dr} \left[r^2 \left(\frac{V^2(r)}{r} - a_D(r) - \frac{V_{HI}^2(r)}{r} \right) \right] \quad (26)$$

3) Baryonic Accelerations

Stellar disk:

$$a_D(r) = \frac{V_{\text{disk}}^2(r)}{r} \quad (27)$$

Gas disk:

$$a_{HI}(r) = \frac{V_{\text{gas}}^2(r)}{r} \quad (28)$$

4) Numerical derivative

Define [4]:

$$q(r) = r^2 \left(\frac{V^2(r)}{r} - a_D(r) - \frac{V_{HI}^2(r)}{r} \right) \quad (29)$$

Then:

$$\rho_{DM}(r_i) \approx \frac{1}{4\pi G r_i^2} \left. \frac{dq}{dr} \right|_{r_i} \quad (30)$$

5) NFW profile

Density profile:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (31)$$

Enclosed mass:

$$M_{\text{NFW}}(< r) = 4\pi\rho_s r_s^3 \left[\ln(1+x) - \frac{x}{1+x} \right], \quad x = \frac{r}{r_s} \quad (32)$$

Relations between parameters:

$$R_{200} = \left(\frac{3M_{200}}{4\pi \cdot 200 \cdot \rho_{\text{crit}}} \right)^{1/3} \quad (33)$$

$$r_s = \frac{R_{200}}{c} \quad (34)$$

$$\rho_s = \frac{M_{200}}{4\pi r_s^3 \left[\ln(1+c) - \frac{c}{1+c} \right]} \quad (35)$$

6) Observational matching

- Choose an external matching radius r_{match} (NGC 3198: $r_{\text{match}} \approx 44$ kpc).
- Compute $\rho_{\text{DM}}(r_{\text{match}})$ using Eq. (30).
- Determine M_{200} from the condition:

$$\rho_{\text{NFW}}(r_{\text{match}}; M_{200}) = \rho_{\text{DM}}(r_{\text{match}}) \quad (36)$$

Then recover:

$$c, \quad r_s, \quad \rho_s$$

from the parameter relations above.

1.2 Velocity distribution in galaxy clusters

- **Objectives:** Help students infer the existence of dark matter in galaxy clusters using classical mechanics and virial theorem. Students will apply the virial theorem to estimate the total mass of a galaxy cluster from the kinetic and potential energy of its galaxies.
- **Dataset Coma cluster (SDSS)** : objid (galaxy ID), ra (right ascension, α), dec (declination, δ), modelmag (apparent magnitude in r band, m_r), modelmagerr, extinction, redshift (z), zErr (redshift error) [10, 1].
- **Dataset Abell Clusters:** Cluster (name), ID (galaxy ID), RAdeg (right ascension in degrees), DEdeg (declination in degrees), RV (radial velocity in km/s), e_RV (velocity error in km/s), q_RV (quality flag), bmag (apparent magnitude in B band). [11]

1. Velocity of each galaxy from redshift (data):

$$v_i = c z_i \quad (37)$$

2. **Number of observed galaxies:** N

3. **Velocity mean:**

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i \quad (38)$$

4. **Observed velocity dispersion:**

$$\sigma_{\text{obs}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (39)$$

5. **Comoving distance:**

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}} \quad (40)$$

6. **Angular diameter distance:**

$$D_A(z) = \frac{\chi(z)}{1+z} \quad (41)$$

7. **Search the cluster centers** from BCG or median coordinates:

$$(\alpha_c, \delta_c) = (\alpha_{\text{BCG}}, \delta_{\text{BCG}}) \quad \text{or} \quad (\text{median}(\alpha), \text{median}(\delta))$$

8. **Angular separation:**

$$\theta = \arccos(\sin \delta_1 \sin \delta_2 + \cos \delta_1 \cos \delta_2 \cos(\alpha_1 - \alpha_2)) \quad (42)$$

9. **Projected radius:**

$$r_{\text{proj},i} = \theta_i D_{A,\text{kpc}} \quad (43)$$

10. **Critical density:**

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} \quad (44)$$

11. **Virial relations:**

$$\sigma_{\text{obs}}^2 = \frac{GM_{200}}{3R_{200}} \quad (45)$$

$$M_{200} = \frac{4}{3}\pi 200 \rho_{\text{crit}} R_{200}^3$$

Dark Matter (NFW) Profile

12. **Concentration (fit relation):**

$$c = A \left(\frac{M_{200}}{M_{\text{pivot}}} \right)^B (1+z)^C \quad (46)$$

13. **NFW density factor:**

$$\delta_c(c) = \frac{200}{3} \frac{c^3}{\ln(1+c) - c/(1+c)} \quad (47)$$

14. **Characteristic radius:**

$$r_{200} = \left(\frac{3M_{200}}{4\pi 200 \rho_{\text{crit}}} \right)^{1/3} \quad (48)$$

15. **Characteristic density and scale radius:**

$$\rho_s = \delta_c(c) \rho_{\text{crit}}, \quad r_s = \frac{r_{200}}{c}$$

16. **Dark matter mass (NFW):**

$$M_{\text{NFW}}(r_{\text{proj},i}) = M_{200} \frac{\ln(1+x_i) - x_i/(1+x_i)}{\ln(1+c) - c/(1+c)}, \quad x_i = \frac{r_{\text{proj},i}}{r_s} \quad (49)$$

Luminous Mass

17. **Luminosity distance (small z):**

$$D_L \approx \frac{c z_{\text{cluster}}}{H_0}, \quad z_{\text{cluster}} = \text{median}(z_i) \quad (50)$$

18. **Distance modulus:**

$$\text{distmod} = 5 \log_{10}(D_L) - 5 \quad (51)$$

19. **Absolute magnitude:**

$$M_r = m_r - A_r - \left[5 \log_{10} \left(\frac{D_L}{10 \text{ pc}} \right) \right] \quad (52)$$

20. **Luminosity:**

$$L_r = 10^{0.4(M_{r\odot} - M_r)} \quad (53)$$

21. **Stellar/baryonic mass per galaxy:**

$$M_{\text{bar},i} = (M/L) L_r, \quad M/L = 2 \quad (54)$$

22. **Total stellar mass:**

$$M_{\text{bar}} = \sum_{i=1}^N M_{\text{bar},i} \quad (55)$$

23. **Characteristic baryonic radius:**

$$r_{\text{bar}} = \left(\frac{3M_{\text{bar}}}{4\pi 200 \rho_{\text{crit}}} \right)^{1/3} \quad (56)$$

24. **Gas mass [2]:**

$$M_{\text{gas}} = M_{500} \cdot 0.093 \left(\frac{M_{500}}{2 \cdot 10^{14}/h} \right)^{0.21} \quad (57)$$

(where $M_{500} \approx 0.7 \cdot M_{\text{tot}}$ and $h = 0.7$)

25. **Total Baryonic Mass:**

$$M_{\text{lum}} = M_{\text{bar}} + M_{\text{gas}} \quad (58)$$

26. **Velocity dispersion (baryons only):**

$$\sigma_{\text{bar}} = \sqrt{\frac{GM_{\text{bar}}}{3r_{\text{bar}}}} \quad (59)$$

Velocity Dispersion and Simulation

25. **Total simulated mass:**

$$M_{\text{tot}} = M_{\text{bar}} + f M_{200} \quad (60)$$

26. **Total simulated velocity dispersion:**

$$\sigma_{\text{sim}} = \sqrt{\frac{GM_{\text{tot}}}{3r_{200}}} \quad (61)$$

27. **Observed histogram:**

$$H_{\text{obs}} = \text{hist}(v_{\text{obs}})$$

28. **Generate normal distribution of simulated velocities:**

$$\sigma_{\text{tot}} = \text{Normal}(\mu = \sigma_{\text{sim}}, \sigma = \sigma_{\text{obs}}, N)$$

29. **Simulated histogram:**

$$H_{\text{sim}} = \text{hist}(\sigma_{\text{tot}})$$

Data visualization

- Plot histograms: number of galaxies on Y-axis, velocity on X-axis.
- Curves: observations (blue), simulated (green), baryonic (red).

Cluster Mass via Virial Theorem

1. **Angular Separation ($\Delta\theta$)** from right ascension (α) and declination (δ) data (converted to radians, where c denotes the cluster center and i the individual galaxy):

$$\Delta\theta = \arccos[\sin(\delta_c)\sin(\delta_i) + \cos(\delta_c)\cos(\delta_i)\cos(\alpha_c - \alpha_i)] \quad (62)$$

2. **Projected Radius** (R_P) from angular separation, assuming a distance $D = 100$ Mpc (radius converted to kpc):

$$R_P = D \cdot \Delta\theta \quad (63)$$

3. **Observed Velocity** (V_{obs}) from redshift data (or radial velocity RV):

$$V_{\text{obs}} = c \cdot z \quad \text{or} \quad V_{\text{obs}} = \text{RV} \quad (64)$$

4. **1-D Velocity Dispersion** (σ_{1D}) (observation):

$$\sigma_{1D} = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i - \bar{V})^2} \quad (65)$$

5. **3-D Velocity Dispersion** (σ_{3D}) (assuming isotropy):

$$\sigma_{3D} = 3 \sigma_{1D} \quad (66)$$

Baryonic components

6. **Convert Distance** D (in Mpc) to pc and compute the **Distance Modulus** (μ):

$$\mu = 5 \log_{10} \left(\frac{D_{\text{pc}}}{10 \text{ pc}} \right) \quad (67)$$

7. **Absolute Magnitude** (M) using apparent magnitude ($m = \text{bmag}$):

$$M = m - \mu \quad (68)$$

8. **Luminosity** (L) of each galaxy from absolute magnitude (M):

$$L = 10^{-0.4(M - M_{\odot})} \quad (69)$$

9. **Total Luminosity of the cluster** (L_{total}):

$$L_{\text{total}} = \sum_{i=1}^N L_i \quad (70)$$

10. **Stellar Mass** (M_{lum}) from luminosity (Υ is the mass-to-light ratio):

$$M_{\text{lum}} = \Upsilon \cdot L_{\text{total}} \quad (71)$$

11. **Gas mass:**

$$M_{\text{gas}} = M_{500} \cdot 0.093 \left(\frac{M_{500}}{2 \cdot 10^{14}/h} \right)^{0.21} \quad (72)$$

(where $M_{500} \approx 0.7 \cdot M_{\text{tot}}$ and $h = 0.7$)

12. **Total Baryonic Mass:**

$$M_{\text{lum}} = M_{\text{bar}} + M_{\text{gas}} \quad (73)$$

Virial Theorem Derivation

11. **Gravitational Potential Energy** (light particle m orbiting heavy particle M):

$$U = -\frac{GMm}{r} \quad (74)$$

12. **Gravitational Force:**

$$F_G = \frac{GMm}{r^2} \quad (75)$$

13. **Centripetal Force:**

$$F_c = \frac{mv^2}{r} \quad (76)$$

14. **Equating Forces** ($F_G = F_c$) to find v^2 :

$$v^2 = \frac{GM}{r} \quad (77)$$

15. **Kinetic Energy** (K , neglecting heavy particle's kinetic energy):

$$K = \frac{1}{2}mv^2 \quad (78)$$

16. **Replacing v^2 in K :**

$$K = \frac{1}{2}m \left(\frac{GM}{r} \right) \quad (79)$$

17. **Replacing Potential Energy U in K :**

$$K = -\frac{1}{2}U \quad (80)$$

18. **Virial Theorem:**

$$2 \langle K \rangle + \langle U \rangle = 0 \quad (81)$$

Cluster Virial Mass and Density

- **Context:** In a cluster, the total potential energy is the sum of all pairwise interactions, and the total kinetic energy is the sum of the kinetic energies of all galaxies. The Virial Theorem

relates these totals. For an isotropic and stable system, the velocity is replaced by the velocity dispersion (σ).

19. **Total Virial Mass of the cluster** (M_{Virial}) (using σ_{3D} and characteristic radius R):

$$M_{\text{Virial}} = \frac{3\sigma_{1D}^2 R}{G} \quad (82)$$

20. **Total Volume** (approximated as a sphere):

$$\text{Volume} = \frac{4}{3}\pi R^3 \quad (83)$$

21. **Luminous Density** (ρ_{lum}):

$$\rho_{\text{lum}} = \frac{M_{\text{lum}}}{\text{Volume}} \quad (84)$$

22. **Total Density** (ρ_{total}):

$$\rho_{\text{total}} = \frac{M_{\text{total}}}{\text{Volume}} \quad (85)$$

Data visualization

- **X-Axis:** Radius (R).
- **Y-Axis (Quantities to Plot):**
 1. ρ_{lum} (Luminous Density, Red)
 2. ρ_{total} (Total Density, Data Blue)
 3. V_{sim} (Total Simulated Velocity, Red)
 4. V_{obs} (Observed Velocity, Blue data points)

References

- [1] Mert Alkan. *Coma Cluster Dataset*. <https://www.kaggle.com/datasets/mertalkan98/coma-cluster>. 2021.
- [2] S. Giodini et al. “Stellar and Total Baryon Mass Fractions in Groups and Clusters Since Redshift 1”. In: *The Astrophysical Journal* 703.1 (2009), pp. 982–993. DOI: 10.1088/0004-637X/703/1/982. URL: <https://doi.org/10.1088/0004-637X/703/1/982>.
- [3] E. V. Karukes and P. Salucci. “The Universal Rotation Curve of Dwarf Disk Galaxies”. In: *Monthly Notices of the Royal Astronomical Society* 465 (2016), pp. 4703–4722.
- [4] E. V. Karukes, P. Salucci, and G. Gentile. “The Dark Matter Distribution in the Spiral NGC 3198 out to 0.22 R”. In: *Monthly Notices of the Royal Astronomical Society* (2015).

- [5] Federico Lelli, Stacy S. McGaugh, and James M. Schombert. “SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves”. In: *aj* 152.6, 157 (Dec. 2016). <https://astroweb.cwru.edu/SPARC/>, p. 157. DOI: 10.3847/0004-6256/152/6/157. arXiv: 1606.09251 [astro-ph.GA].
- [6] P. Li et al. “A Constant Characteristic Volume Density of Dark Matter Haloes from SPARC Rotation Curves”. In: *Astronomy Astrophysics* 623 (2019), A70.
- [7] J. F. Navarro, C. S. Frenk, and S. D. M. White. “The Structure of Cold Dark Matter Halos”. In: *The Astrophysical Journal* 462 (1996), p. 563.
- [8] J. F. Navarro, C. S. Frenk, and S. D. M. White. “A Universal Density Profile from Hierarchical Clustering”. In: *The Astrophysical Journal* 490 (1997), p. 493.
- [9] P. Salucci et al. “The Universal Rotation Curve of Spiral Galaxies. II: The Dark Matter Distribution out to the Virial Radius”. In: *Monthly Notices of the Royal Astronomical Society* 378 (2007), pp. 41–47.
- [10] SDSS Collaboration. *Sloan Digital Sky Survey: Science and Data-Release Publications*. Sloan Digital Sky Survey. 2023. URL: <https://www.sdss.org/science/data-release-publications> (visited on 11/13/2026).
- [11] M.J. Way et al. “Redshifts in the Southern Abell Redshift Survey Clusters. I. The Data”. In: *aj* 130.5 (2005), pp. 2012–2018. DOI: 10.1086/462418. arXiv: astro-ph/0509736 [astro-ph].